Sections: B,C & D

Linear Algebra (MT-1004)

Homework#38

Instructor: Muhammad Ali

Question 1

Square Root of a Matrix

Let

$$A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}.$$

Theoretical Analysis:

(a) Show that the matrix A is symmetric.

(b) Verify if A is positive definite.

Computation of Square Root:

1. Find the eigenvalues and eigenvectors of A.
2. Using the spectral decomposition $A = Q\Lambda Q^T$, compute the square root of A, denoted as $A^{1/2}$, where $A^{1/2}A^{1/2} = A$.

Question 2

Rayleigh Quotient

Let

$$A = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}$$

be a symmetric matrix.

Definition and Properties:

1. Write the formula for the Rayleigh Quotient $R(x, A)$ of a vector x with respect to the matrix A.

Computation:

(a) Compute the eigenvalues of A and identify $\lambda_{\min}(A)$ and $\lambda_{\max}(A)$.

(b) For $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, compute the Rayleigh Quotient $R(x, A)$.